

Research paper

The impact of low-carbon city pilot on carbon emissions of high-polluting enterprises – Based on financing constraint perspective

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ABSTRACT

As the major contributor to climate change, enterprises in high-polluting industries have a large deal of potential to reduce their carbon emissions, and the efficacy of that reduction is correlated with the nation's overall environmental pollution and climate change. As a comprehensive environmental regulation policy, whether the low-carbon city pilot (LCCP) can promote the carbon emission reduction of high-polluting enterprises remains to be verified. Based on the data of all high-polluting listed companies in China during the period of 2010–2020, this study evaluates the implementation effect of the LCCP through the multi-period difference-in-differences model, and conducts a series of robustness tests. In addition, the implementation path, heterogeneity and industry spillover effects of the policy are also studied. The study shows that the LCCP policy significantly reduces the growth rate of carbon emissions of enterprises in high-polluting industries by 1.06 percentage points on average. Financing constraints play a negative moderating role between the LCCP and the growth rate of carbon emissions. The mechanism test demonstrates that the LCCP reduces the carbon emission growth rate of enterprises in high-polluting industries through enterprise technological innovation and increasing environmental protection investment level. Further analysis can also find that the LCCP has a more obvious emission reduction effect on state-owned enterprises and enterprises in areas with lax environmental law enforcement. And there is an economic spillover effect but no industrial spillover effect. In view of this, local governments should play a guiding role in the policy, continue to expand the scale of the LCCP and focus on non-state-owned enterprises to promote the green transformation of high-polluting enterprises. This study offers significant theoretical and practical reference value for the further implementation of environmental regulation in the field of environmental governance and the reduction of enterprise carbon emissions.

1. Introduction

Climate change is a joint challenge faced by all humankind. The Earth's climate system is changing dramatically due to rising greenhouse gas emissions, which presents serious hazards to our society, economy, and environment. The urgent need to reduce greenhouse gas emissions, build resilience, and adapt to climate change are the main topics of discussion at the 27th United Nations Climate Change Conference in November 2022. The importance of addressing climate change has increased around the world. China is one of the world's top emitters of greenhouse gases. During its early years of rapid economic expansion, the country's industrialization process accelerated and its energy consumption increased significantly, resulting in high carbon dioxide emissions. The International Energy Agency's "Global Clean Energy Investment Report in 2021" highlights the importance of

increasing investment in renewable energy and accelerating clean energy development, and points out that reducing fossil fuel use is essential to achieving carbon emission reduction. 5.41 billion tons of standard coals were consumed in China in 2022, making about 56.2 % of the country's total energy consumption. China continues to be a major energy consumer and producer of carbon dioxide, with emissions of 1147.7 million tons, a decrease of 23 million tons from 2021.

Policies, laws and regulations drive countries towards sustainable development by regulating and guiding social behavior. At present, at the international level, relevant policies have been developed and implemented to reduce greenhouse gas emissions and promote the national transition to low carbon (Eskander and Fankhauser, 2020). Italian legislation introduces the concept of "producer and marketer", which greatly promotes the transformation of energy to low carbon and clean through the promotion of photovoltaic energy systems. The

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development and utilization of renewable energy are of great significance to the realization of national sustainable development (Lucchi et al., 2023). The Paris Agreement aims to limit global warming to less than 2 degrees Celsius through global cooperation and strives to limit the temperature rise to 1.5 degrees Celsius. In response, Brazil developed INDCs to encourage the use of wind, solar and biofuels, fulfilling plans to reduce greenhouse gases (Lima et al., 2020). In order to reduce the impact of municipal solid waste on greenhouse gas emissions, Malaysia promulgated the Kyoto Protocol, which finally proved that such management is effective for greenhouse gas emission reduction (Devadoss et al., 2021). China has made commitments and taken steps to combat climate change and environmental pollution. Since the "dual-carbon" goal was proposed, China has formulated a number of policies to reduce emissions and save energy. As the main body of carbon emission, the carbon reduction and upgrading of enterprises in high-polluting industries is essential for improving environmental quality, reducing greenhouse gas emissions, and realizing sustainable economic development. In February 2022, a document issued by the National Development and Reform Commission pointed out that through guiding transformation technology, strengthening technological research, accelerating the elimination of backward equipment, and other measures to urge high-energy enterprises to implement practical steps to cut back on energy consumption and carbon emissions. Meanwhile, through the expansion of renewable energy, improving energy efficiency, and promoting cleaner production measures, China is committed to fostering economic structure adjustment, accelerating the transition to clean energy and low-carbon technologies, and fostering an environment that benefits both economic development and environmental preservation. In this context, has the "top-down" formal environmental regulation, such as the LCCP policy enacted by the government, achieved the desired results for enterprises in high-polluting industries that are major emitters? This merits more investigation. An in-depth study of the key factors influencing carbon emission reduction by enterprises, the realization of "carbon peak" and "carbon neutrality", as well as the crises and challenges posed by climate change, are all extremely important from both an academic and practical standpoint.

At present, the literature on the environmental impact of the LCCP mostly focuses on the macro level of provincial and municipal level, and relatively few studies involve the micro-enterprise level. This paper fills the research gap in this aspect. The majority of literature mainly examines the impacts of the LCCP policy on enterprises from the economic effects, including enterprise total factor productivity (Wen et al., 2023; Zhao et al., 2021), financial performance (Sun et al., 2022; Wang et al., 2023), green technology innovation (Xiao et al., 2023; Xu and Cui, 2020) and so on. In the meantime, the influence of the policy on the urban level is also studied from environmental governance effects (Huo et al., 2022; Wen et al., 2022; Yu and Zhang, 2021). Scholars have confirmed that the LCCP policy reduces carbon emissions at the macro level, but there is still a dearth of research on how the policy affects emissions at the micro-enterprise level. At the same time, from the background of the era of carbon emission reduction of high-polluting enterprises, few scholars have conducted in-depth studies on the impact of financing constraints on the relationship between the two. Existing literature on the LCCP mostly focuses on economic benefits or environmental effects unilaterally. This paper further comprehensively analyzes the economic effects and industrial spillover effects of the LCCP. As the main contributor to climate change, enterprises in high-polluting industries have enormous potential to reduce carbon emissions and ought to be the main target of the LCCP policy (He, Jing, and Chen, 2023). So, does the LCCP policy's implementation effectively mitigate carbon emissions at the enterprise level and slow down the rate of carbon emissions of enterprises in high-polluting industries? If so, what is the mechanism by which enterprises reduce their carbon emissions? Does the implementation of the LCCP work differently for enterprises of different ownership? Are there economic effects and industrial spillover effects? The responses to the aforementioned queries

will not only enrich existing research and promote the green transformation of enterprises but also provide reference value and policy recommendations for China to realize the "dual carbon" target.

Based on the sample data of high-polluting enterprises in China from 2010 to 2020, this study empirically examines the impact of the LCCP on carbon emission reduction of high-polluting enterprises by using the difference-in-differences model. By comparing the data of carbon emission growth rate before and after the implementation of the policy, we investigate how the LCCP affects the carbon emission growth rate of enterprises. It also examines whether there are differences in the emission reduction effects of the LCCP on different enterprise ownerships, different enterprise ages, different enterprise sizes, and the strength of regional environmental enforcement. The impact of the policy implementation on the economic efficiency of enterprises and the environmental effect of the neighboring areas are further examined.

The research objective of this paper is to analyze the effect of the LCCP on carbon emission reduction of high-polluting enterprises through the difference-in-differences method. Combined with the Porter hypothesis, pollution paradise hypothesis and factor endowment hypothesis, this paper analyzes the mechanism in depth, summarizes the effect of the policy implementation, and finally puts forward feasible suggestions for the further implementation of the LCCP. Compared with the previous literature, this study primarily contributes in the following three aspects: First, it examines the impact mechanism of the LCCP on carbon emission reduction of enterprises from a micro perspective, and tests the mechanism role of technological innovation and environmental protection investment level of enterprises; Second, from the background of the era of carbon emission reduction of high-polluting enterprises, it helps to supplement the related inquiry of financing constraints in carbon emission reduction of enterprises in the LCCP. Third, further analysis of its economic and industrial spillover effects, and the study findings will help to more deeply understand the driving role of the LCCP on high-polluting enterprises. Relevant research and findings provide experience for China's carbon emission reduction and also have certain reference significance for other developing countries.

2. Literature review

2.1. Research progress of environmental regulation on carbon emission reduction

In recent years, the world's rapid economic development has been accompanied by increasing resource scarcity, climate change, and environmental pollution. Environmental regulation is acknowledged by all nations as one of the most efficient methods to lower carbon emissions. Many scholars have carried out how environmental regulations affect the reduction of carbon emissions, but the current research has not yet come to a unified conclusion.

Some studies have concluded that environmental regulation contributes positively to the decrease of carbon emissions. Zhang, Li, and Wang (2023) investigated how market-oriented environmental regulation policies affect enterprises' carbon emissions using energy-consuming right trading as an example. They discovered that environmental regulation can greatly lower carbon emissions by promoting green technological innovation. Xu and Li (2024) studied the environmental protection tax law using data from listed companies and found that the green tax reform realizes carbon emission reduction of enterprises and promotes low-carbon development transition by increasing environmental protection investment and promoting green innovation. Shen, Pan, and Feng (2023) used data from 20 provinces to study informal environmental regulations that have received little attention and discovered that informal environmental regulations have a significant contribution to regional carbon emissions. Neves et al. (2020) examined the impact of environmental regulations on CO₂ emissions in the short and long term using annual data from 27 EU countries. The study found that support for renewable energy policies

led to reductions in CO₂ emissions in both the short and long term, and also showed that the EU is attracting high-quality and innovative investments.

Some literature argues that environmental regulation may have a detrimental effect on carbon emissions. Huang and Song (2023) studied the tendency of industrial land transfer structures by using the super-efficiency SBM-DEA model and discovered a negative association between carbon emission efficiency and environmental regulation. Dabuo et al. (2023) constructed a spatial Durbin model to study the influence of spatial spillover effects of environmental regulation on carbon emissions. They discovered that energy consumption, R&D intensity, and environmental regulation all work together to increase carbon emissions and that the relationship between energy consumption and carbon emissions is strongly regulated by strict environmental regulatory measures.

Some other scholars believe there is a threshold between environmental regulation and carbon emissions, and the relationship between the two changes when the threshold is exceeded. Jiang and Ma (2021) found that there is a "U-shaped" relationship between environmental regulation and the carbon emission network in the Organization for Economic Cooperation and Development by applying the social network analysis. When the environmental regulation reaches the threshold, the innovation compensation effect starts to appear. Wang et al. (2021) studied the impact of environmental regulation on environmental pollution based on the Nested Dixit-Stiglitz model and found that the relationship between environmental regulation and pollution emission shows a "U" shape and skill premium plays a negative role in the middle. Hu and Wang (2020) discovered through the calculation and analysis of the change in carbon productivity that the impact of environmental regulation on carbon production has a threshold. When environmental regulation increases in the later stage, the effect on carbon productivity shifts from being negative to positive. After gathering carbon emission data from 30 provinces, Ye et al. (2023) discovered that formal environmental regulation has a "U-shaped" relationship with carbon emission efficiency, while informal environmental regulation has an inverted "U-shaped" relationship with carbon emission efficiency.

2.2. Relevant research on low-carbon city pilot

The study on the impact of the LCCP policy has long been a hot topic in academia. The existing research has mainly explored the environmental and economic aspects to examine the impacts of policy effects. First, existing scholars generally agree that the LCCP will have a favorable environmental effect and lessen carbon emissions. Wen et al. (2022) used SBM to measure the carbon emission efficiency of cities and found that the LCCP significantly improved the carbon emission efficiency of cities. The policy had great regional heterogeneity, and the effect of carbon efficiency improvement was especially obvious in the third-tier cities and resource-based cities. Zhang et al. (2022) analyzed the impact of the LCCP on Central and Eastern Europe by taking advantage of the differences of the LCCP implemented between cities and within their life cycle. The study found that pilot cities' CEE region increased their carbon emission efficiency by 2.04 % compared to non-pilot cities, and the effect strengthened over time. The effect of the policy implementation is more obvious in the east, west, non-mineral resource-based and high innovation level cities. Zeng et al. (2023) used the spatial difference method to quantitatively examine the carbon emission reduction effect of the LCCP and found that under the influence of urban environmental governance, industrial structure optimization and urban innovation level improvement, the LCCP significantly reduced the carbon intensity of the pilot city by 0.13 %. At the same time, there is a spatial spillover effect, which reduces the carbon intensity of neighboring cities by 0.9 %. Lyu et al. (2023) used batch decomposition for the first time to assess the effects of three batches of the LCCP and the differences between batches. The adoption of the legislation significantly reduced both the overall and per-person carbon

emissions, according to the research. The first and second batches of the LCCP had considerable effects, while the third batch of the implementation effect was not significant. Second, due to differences in the measurement indicators and the selection of pilot batches, the existing documentation on the economic benefits of the LCCP is inconsistent in its conclusions on economic benefits. Yao and Shen (2021) conducted a joint analysis of the LCCP on air quality and labor productivity to explore the applicability of the policies to economic development models. The study found that the LCCP improves air quality but neglects economic development. Cheng et al. (2019) used the second batch of pilot cities and discovered that the LCCP contributes positively to the growth of the green economy in pilot areas, and it has economies of scale. The effect is more noticeable in larger cities. Wang and She (2020) measured the economic benefits of the first two pilot cities by the level of green growth and discovered that the pilot policy has greatly positive effects on economic growth, especially for the eastern region and large-scale cities. Fan et al. (2021) used the data envelopment analysis approach to measure the low carbon economy and found that the LCCP significantly boosted the low carbon economy of the city and increased it over time. The enhancement was more pronounced in the central and western regions and in large cities without strict environmental protection plans.

After summarizing and reviewing the body of research, it is found that there are still vacancies in the research field of the LCCP. First, the majority of studies on the impact of LCCP on reducing carbon emissions have concentrated on the nation, provinces, and industries at large. Few have examined how policies affect carbon emissions from the viewpoint of the microenterprise. There is a lack of research on the implementation of policies for high-polluting enterprises, which is the main body of carbon emission reduction, and it needs to be further explored. Second, from the background of the era of carbon emission reduction of high-polluting enterprises, few academics have studied in depth the impact of financing constraints on the relationship between the two. Given this, based on the data of all listed enterprises in China's high-polluting industries from 2010 to 2020, this paper uses the multi-period difference-in-differences model to investigate the impact of the LCCP on the growth rate of carbon emissions of enterprises in high-polluting industries. It further explores the moderating effect of financing constraints on the relationship between the two and examines the mechanism of enterprise technological innovation and environmental protection investment levels.

3. Theoretical analysis and research hypothesis

The LCCP policy is more flexible and scientifically designed than traditional environmental policies (Zheng and Chang, 2023), with pilot cities choosing suitable measures and targets to achieve optimal carbon emission reduction based on the features, stages of development, and resource status of their respective cities. The policy design allows cities to continuously adjust and optimize their attempts to respond to changes in the environment and demand. The city's carbon emission profile, energy structure, economic development, and other aspects were taken into account when formulating the policy, and low-carbon targets and measures were established in a scientific manner. According to the review of the LCCP policy, it is known that the implementation includes the following parts: setting up a statistical and management system for greenhouse gas emission data, establishing supporting policies for low-carbon development, advocating a low-carbon green life for the whole people, establishing a low-carbon industrial system, creating a responsibility system for controlling the target of greenhouse gas emission and establishing a system of assessment for controlling the target of greenhouse gas emission. The implementation of relevant policies in the pilot regions based on the above requirements is not only crucial for combating climate change and achieving carbon emission reduction but also for fostering the growth of the green economy, improving urban sustainability, enhancing competitiveness, and helping the achievement

of global sustainable development goals. Steel, pharmaceutical, textile, mining, metallurgy, coal, chemical industry, and the other 16 categories of industries are defined as high-polluting industries. They usually have high carbon emission intensity and massive potential for emission reduction, so their active participation in carbon emission reduction is the key to realizing the LCCP policy. This study provides the following hypothesis in light of this:

Hypothesis 1. : The implementation of the LCCP policy helps reduce the growth rate of carbon emissions of enterprises in high-polluting industries.

To come true the low-carbon goal, China has implemented the LCCP policy, which stimulates enterprises and research institutions to engage in technological innovation, that is, realizing the "Porter's hypothesis". This theory postulates that effective environmental regulation has an "innovation compensation" impact and puts pressure on enterprises to innovate (Porter, 1996). Environmental regulation raises the cost of environmental compliance, but over time, enterprises will increase their investment in innovation and cut back on pollution emissions to enhance productivity and keep up with the competition (Xiang and Zhao, 2023). Environmental regulation requires enterprises to meet higher environmental standards, and this pressure stimulates the motivation of enterprise innovation. Enterprises enhance production technology and resource efficiency in search of novel ways to lower carbon emissions. These adjustments assist enterprises in meeting consumer demand for environmentally friendly goods and services while giving them a competitive edge in markets where environmental awareness is growing. According to the body of existing research, the LCCP policy encourages green technology innovation among enterprises through innovative incentives like tax breaks and enhanced R&D subsidies (Li and Wang, 2023). At the same time, it can also enhance the technological innovation of enterprises by increasing the promotion of industrial structure upgrading, promoting population flow, and improving the public's attention to environmental protection (Deng et al., 2022). Accordingly, the research hypothesis presented in this research is as follows::

Hypothesis 2a. : The LCCP policy reduces the growth rate of carbon emissions of enterprises in high-polluting industries through enterprise technological innovation.

The Pollution Heaven Hypothesis and the Factor Endowment Hypothesis are two theories that are used in this research to analyze how environmental regulations affect enterprises' investments in environmental protection. The Pollution Heaven Hypothesis states that enterprises in areas with more stringent environmental laws will face higher production costs. As a result, enterprises may decide to relocate their operations to areas with less stringent laws. Strict environmental regulation may result in lower levels of investment in the environment, particularly in the context of globalization, when enterprises often prefer to produce in less regulated areas to save money. According to the Factor Endowment hypothesis, enterprises will actively respond to environmental regulations and amplify their expenditures on environmental protection when the expense is less than the reward. On the contrary, if enterprises cannot benefit from environmental protection investments, they will be discouraged from investing in environmental protection. Despite having different points of view, both hypotheses contend that environmental regulation affects the amount of money that businesses invest in environmental protection.

Combining the above two hypotheses and the scientific and normative nature of the LCCP policy, this paper proposes that under the impetus of the LCCP policy enterprises will lower the growth rate of carbon emissions in high-polluting industries by upgrading the level of environmental protection investment. On the one hand, as the pilot cities gradually tighten environmental enforcement, enterprises must prioritize environmental governance and respond to environmental rigidity constraints by increasing environmental protection investment

when the environmental protection tax and environmental protection penalty paid by enterprises are significantly greater than the required environmental protection investment (Gray and Deily, 1996). To cut down on energy consumption and trash emissions, enterprises have extensively invested in introducing more advanced and cleaner production processes and facilities. The application of environmental protection technology aids enterprises in becoming more productive and resource-efficient, which lowers the pace of growth in carbon emissions. On the other hand, the government encourages enterprises to invest in the usage of clean energy by offering financial assistance, such as direct subsidies or low-interest loans to enterprises to lower their environmental investment costs. Enterprises are investing in the use of clean energy, such as solar, wind, and geothermal energy, to replace traditional fossil fuels and reduce their dependence on finite resources. They also invest in developing and promoting environmentally friendly products and materials, including using renewable materials and promoting environmental improvements in product design. Furthermore, pressure from society and investors has pushed enterprises to invest in environmental protection. Enterprises have been driven to raise their environmental investments and improve their reputation for sustainable management as a result of the growing public awareness of environmental challenges and the demands of investors to assess enterprise ESG indicators. Therefore, the enterprise can formulate its environmental protection investment strategy and incorporate it into the long-term planning and strategy in accordance with the characteristics of the industry and the operation of the enterprise. Drawing from the above theoretical analysis, the following hypothesis is proposed in this paper:

Hypothesis 2b. : The LCCP policy reduces the growth rate of carbon emissions of enterprises in high-polluting industries by upgrading the level of enterprises' environmental protection investment.

Extensive development mode, unreasonable industrial structure, and energy structure have caused a large amount of carbon emissions from high-polluting enterprises. Pollution control requires enterprises to invest a lot of manpower and material resources. It is not only difficult for enterprises to rely on their own funds to invest but also may affect the normal production and operation of enterprises. At the same time, too much use of enterprises' own funds for pollution control is not conducive to improving the enthusiasm of enterprises for pollution control. Discussing the financing constraints faced by high-polluting enterprises can be understood from two aspects. On the one hand, most high-polluting enterprises have high environmental risks, and financial institutions usually consider their riskiness and increase their lending rates when considering lending. On the other hand, in the face of environmental pressure and governance requirements, high-polluting enterprises may need to make investments in equipment upgrades, technological improvement, and emission control. These investments may adversely affect the cash flow and profitability of enterprises, increase the uncertainty of financial institutions about the future profitability of enterprises, and reduce the feasibility and cost-effectiveness of financing. If the financing difficulties and expensive problems faced by enterprises are not effectively solved, the lack of sufficient financial support for corporate pollution control will lead to greater resistance to corporate carbon emission reduction. Therefore, the following hypothesis is proposed:

Hypothesis 3. Financing constraints have a negative effect on high-polluting enterprises. The higher the degree of financing constraints faced by enterprises, the worse the carbon emission reduction effect of the LCCP.

4. Research design

4.1. Model setting

To test whether the implementation of the LCCP policy has

effectively slowed down the growth of enterprise carbon emissions, this study builds a multi-period difference-in-differences (DID) model (1). The difference-in-differences model is used as a method for assessing the effects of policy implementation, viewing policy as a natural experiment. Multi-period difference-in-differences is used when the research object is affected by the policy impact in different times. First, the net effect due to policy implementation is examined by including a control group in the sample that is not affected by the policy and analyzing it in comparison with the experimental group that is affected by the policy. Second, two differences were made before and after the intervention, and the difference between the two groups was obtained, representing the relative relationship between the experimental group and the control group before and after the intervention. Finally, the second difference between the two groups was carried out to eliminate the original difference between the experimental group and the control group, and finally the net effect was obtained. The use of the difference-in-differences model to evaluate policy effects can largely avoid the problem of endogeneity. Policies are generally exogenous with respect to micro-subjects, and thus the problem of reverse causation does not exist.

The independent variable acts on the dependent variable, but the magnitude and direction of the action intensity are affected by the regulating variable Z . Z is the regulating variable at this time, and the effect is called the regulating effect. In order to investigate the moderating effect of financing constraints on the LCCP and carbon emissions, $\text{time} \times \text{treat} \times \text{FC}$ interaction variable is introduced and model (2) is designed to test Hypothesis 3.

$$R_{it} = \beta_0 + \beta_1 \text{time} \times \text{treat} + \lambda X_{it} + \gamma_i + \mu_i + \varepsilon_{it} \quad (1)$$

$$R_{it} = \beta_0 + \beta_1 \text{time} \times \text{treat} + \beta_2 \text{time} \times \text{treat} \times \text{FC} + \beta_3 \text{FC} + \lambda X_{it} + \gamma_i + \mu_i + \varepsilon_{it} \quad (2)$$

Among them, the explanatory variable R_{it} presents the carbon emission growth rate of enterprise i in the high-polluting industry in year t ; β_0 is a constant term; $\text{time} \times \text{treat}$ indicates the LCCP, if the city where enterprise i is located in year t has implemented the pilot policy, it will take the value of 1. If enterprise i does not belong to the LCCP or belongs to the LCCP but the pilot has not yet begun in the year, then the value will take the value of 0; β_1 indicates the magnitude and direction of the role of the LCCP on the growth rate of carbon emissions of enterprises in high-polluting industries; X_{it} denotes a series of control variables that affect the carbon emission growth rate of enterprise i in year t , including the proportion of independent directors, the shareholding ratio of institutional investors, the combination of two positions, Tobin's Q , the nature of the shareholding, the age of the enterprise, the capital intensity, and government R&D investment; γ_i , μ_i denotes time-fixed effect and individual-fixed effect respectively; ε_{it} is a random error term; FC is the degree of financing constraint; β_2 is an interactive effect.

4.2. Selection of variables

4.2.1. Dependent variable

Drawing on Bolton and Kacperczyk (2021), the growth rate of corporate carbon emissions (R) is selected as the explained variable in this study to measure the effect of corporate carbon emission reduction. According to the calculation of enterprise carbon emissions by Wang et al. (2022), it includes the following four components: emissions from combustion and energy fuel, emissions from the production process, emissions from land use transformation, emissions from solid waste combustion and emissions from sewage treatment. Where C is the enterprise carbon emissions in the base period and ΔC is the difference between enterprise carbon emissions in the current period and the base period.

$$R = \frac{\Delta C}{C} \quad (3)$$

4.2.2. Explanatory variable

The LCCP policy was launched in July 2010, November 2012, and January 2017. This study performs research based on the second and third batches of the LCCP policy since they are complete and have better implementation effects. Considering the accuracy of the data and the operability of this study, this paper sets the starting time of the second and third batches of the LCCP policy as 2012 and 2017, respectively, to study the impact of the LCCP policy on the growth rate of carbon emissions of enterprises in high-polluting industries. The core explanatory variable in this paper is $\text{time} \times \text{treat}$. Time is a time dummy variable, taking 1 for the year of policy implementation and beyond, and 0 for the rest. Treat is a policy dummy variable, taking 1 if it is a pilot city and 0 for the remaining.

4.2.3. Moderating variables

This paper uses financing constraints (FC) as a moderating variable. The FC index focuses on the impact of cash flow within the enterprise on financing demand and evaluates the financing constraints. The larger the FC index, the higher the degree of corporate financing constraints. β_2 is an interactive effect. When $\beta_2 > 0$, it indicates that financing constraints have a weakening effect on the relationship between the LCCP and corporate carbon emission reduction. Conversely, when $\beta_2 < 0$, it strengthens the moderating effect.

4.2.4. Control variables

Referring to Ren et al. (2020) and Cao, et al., (2022), this paper selects the proportion of independent directors (Indep), Tobin's Q (TobinQ), the combination of two positions (Dual), the nature of the shareholding (SOE), the age of the enterprise (FirmAge), the shareholding ratio of institutional investors (INST), the capital intensity (Capint) as control variables at the enterprise level. Drawing on Zhao et al. (2021), government R&D investment (GII) is selected as the control variable at the city level. The definitions are displayed in Table 1.

4.3. Sample selection and data sources

This paper takes all listed enterprises in China's high-polluting industries from 2010 to 2020 as a sample. According to the Guidelines on Environmental Information Disclosure of Listed Enterprises issued by the Ministry of Environmental Protection in 2010, 16 categories of

Table 1
Variable description.

Variable type	Variable name	Variable description
Dependent variable	The growth rate of carbon emissions(R)	Difference between current and base period carbon emissions divided by base period carbon emissions
Explanatory variable	LCCP(time×treat)	The enterprise takes 1 in the year or after the pilot city, otherwise 0
Control variables	Proportion of independent directors (Indep)	Number of independent directors divided by number of directors
	Combination of two positions(Dual)	The chairman of the board and the general manager are the same person as 1, otherwise 0
	Tobin's Q(TobinQ)	The market value of shareholders' equity and liabilities divided by total book assets
	Nature of the shareholding(SOE)	The value is 1 for state-owned enterprises and 0 for others
	Age of the enterprise (FirmAge)	Number of years the company was established
	Shareholding ratio of institutional investors (INST)	Total number of shares held by institutional investors divided by outstanding share capital
	Capital intensity(Capint)	Fixed assets divided by total assets
	Government R&D investment(GII)	Government funding divided by internal expenditure on R&D

industries such as steel, pharmaceutical, textile, mining, metallurgy, coal, and chemical industries are mainly defined as high-polluting industries. The control variables include enterprise and city levels, in which the data of control variables at the city level come from the China Urban Statistical Yearbook (the latest statistical indicator deadline is 2020). The cut-off time of the research samples is set to 2020 to harmonize the cut-off times of the chosen variables. Other data comes from the China Stock Market & Accounting Research (CSMAR) Database.

This study processes the original data in accordance with the following three procedures to minimize the impact of anomalous samples: ① Excluding listed enterprises with "ST" and "*ST" in their stock short names; ② Excluding samples with abnormal financial status or other abnormalities (ST, ST, *ST and S*ST enterprises) and samples of financial listed enterprises; ③ listed enterprises with seriously missing data are excluded. Following the above processing, 2894 observations and 570 listed enterprises comprise the unbalanced panel data. The descriptive statistics of variables are shown in Table 2.

5. Empirical analysis

5.1. Benchmark regression analysis

Table 3 reports the baseline regression results of the impact of the LCCP on the carbon emission growth rate of enterprises in high-polluting industries. Without controlling any influencing variables, the regression outcomes in column (1) of Table 3 show that the estimated coefficient of time×treat is negatively significant at the 10 % level, indicating that the implementation of the LCCP policy is able to reduce the growth rate of carbon emissions of enterprises in high-polluting industries. Due to its characteristics such as the scale and operating conditions of the enterprise, the actual conditions such as the economic development of the city where the enterprise is located will also impact the pollution emissions of the enterprise. As a result, this paper manages the city and enterprise-level influencing factors. At the same time, in order to control the emission reduction changes caused by enterprises themselves and the development of time, this paper adds enterprise and year-fixed effects. The estimated coefficient of time×treat is negatively significant at the 5 % level, and the core conclusions of this paper have not been fundamentally changed, which is displayed in column (2) of Table 3. The coefficient in column (2) is slightly smaller and more significant than in column (1), which indicates that there are factors in the control variables that affect the growth rate of carbon emissions of enterprises in high-polluting industries. Theoretically, the net effect of the LCCP policy on the growth rate of carbon emissions of enterprises in high-polluting industries can be estimated more accurately with the inclusion of control variables and fixed effects. According to the findings in column (2), enterprises in the LCCP areas have a 1.06 percentage point lower growth rate in carbon emissions than those that are not. The above results preliminary provide evidence that Hypothesis 1 is valid.

The interaction regression coefficients in column (3) of Table 3 are positively significant at the 1 % level, which suggests that financing

Table 2
Descriptive statistics.

Variable	Obs	Mean	SD	Min	Max
R	2112	0.0082	0.0453	-0.2627	0.4580
time×treat	2112	0.2744	0.4489	0.0000	1.0000
Indep	2112	0.3677	0.0502	0.2727	0.6000
Dual	2112	0.1745	0.3691	0.0000	1.0000
TobinQ	2112	1.6674	0.9393	0.8024	9.8236
SOE	2112	0.5677	0.4909	0.0000	1.0000
FirmAge	2112	2.9101	0.2964	1.3863	3.5553
INST	2112	0.4573	0.2270	0.0000	0.8867
Capint	2112	2.0368	1.6651	0.1314	27.9946
GII	2112	0.0317	0.0232	0.0007	0.1867

Table 3
Benchmark regression results.

Variable	(1)	(2)	(3)
time×treat	-0.0079* (0.0046)	-0.0106** (0.0047)	-0.0234*** (0.0083)
time×treat×FC			0.0511*** (0.0179)
FC			0.0015 (0.0122)
Indep		0.0210 (0.0357)	0.0497 (0.0418)
Dual		0.0012 (0.0034)	0.0013 (0.0038)
TobinQ		0.0004 (0.0023)	0.0012 (0.0025)
SOE		0.0114 (0.0096)	0.0093 (0.0083)
FirmAge		-0.0125 (0.0302)	0.0027 (0.0341)
INST		0.0084 (0.0094)	0.0067 (0.0101)
Capint		-0.0028 (0.0020)	-0.0030 (0.0022)
GII		0.1740* (0.1013)	0.1936* (0.1063)
Cons	0.0100*** (0.0013)	0.0279 (0.0940)	-0.0269 (0.1056)
Enterprise FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	2112	2112	2112
R ²	0.1450	0.1560	0.1636

Note: *** denotes $p < 0.01$, ** denotes $p < 0.05$, * denotes $p < 0.1$; enterprise-level clustering robust standard errors in parentheses.

constraints play a negative moderating role between the LCCP and the growth rate of carbon emissions. That is, the enhancement of financing constraints of high-polluting enterprises weakens the negative relationship between the LCCP and the growth rate of carbon emissions, and Hypothesis 3 is verified. When enterprises face rising financing constraints, their own funds are limited, and the funds available for them to spend on pollution control and equipment upgrading are reduced, which is not conducive to their carbon emission reduction.

5.2. Robustness test

5.2.1. Parallel trend test

To ensure the reliability of the conclusions, the explanatory variables of the experimental group and the control group should maintain the same trend before employing the multi-period difference-in-differences model to evaluate the policy. Then it is consistent with the parallel trend. This paper uses event analysis to conduct the parallel trend test as shown in Eq. (4):

$$R_{it} = \beta_0 + \sum_{t=-4}^4 Policy_{it} + \lambda X_{it} + \gamma_i + \mu_i + \varepsilon_{it} \quad (4)$$

Where, $Policy_{it}$ is a dummy variable that takes the value of 1 if the city where enterprise i is located has implemented the LCCP policy in year t , otherwise it takes the value of 0. The rest symbols have the same meanings as those in Eq. (1). The time span of the pilot is calculated based on the years of the second and third batch of the LCCP policy, and the data selected in this paper is from 2010 to 2020. This paper summarizes the data of the first 4 years before the policy implementation into the -4th period and summarizes the data of the 4 years after the implementation of the policy into the 4th period.

The findings are displayed in Fig. 1, which supports the parallel trend theory by demonstrating that there was no discernible variation in the growth rate of carbon emissions before the LCCP policy. In addition, implementing the LCCP policy is a continuous and dynamic process. The dynamic analysis of the parallel trend chart displays that the estimated

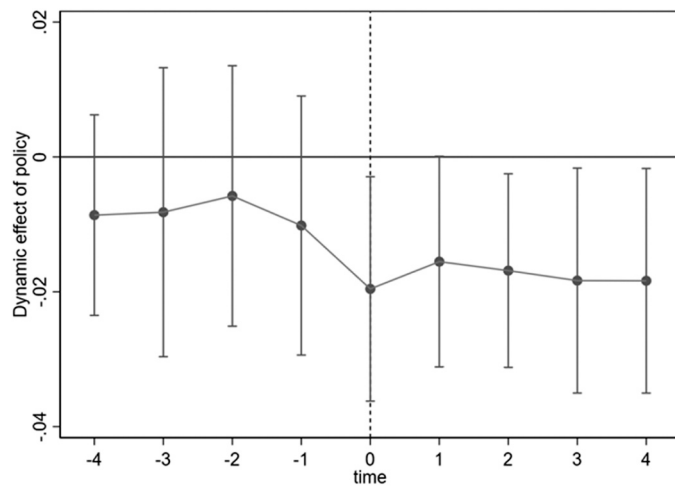


Fig. 1. Parallel trend test.

coefficients are not significant in the four years prior to the implementation of the LCCP policy. At this time, there is no significant difference in the carbon emission growth rate of enterprises in high-polluting industries. Whereas, the policy's dynamic influence starts to manifest in the year of policy implementation and the coefficient falls off the most. The new policy's execution may be strongest just after it's passed, and the immediate results are the best. The growth rate of carbon emissions fluctuates downward in the years that follow, and the significance level remains stable and rising. This suggests that the LCCP policy significantly reduces the growth rate of carbon emissions of enterprises in high-polluting industries and has a certain degree of continuity.

5.2.2. Placebo test

To further test whether the impact of the LCCP policy on enterprises in high-polluting industries is affected by other unobservable factors, this paper uses the substitution test to implement a placebo test. Specifically, referring to the practice of Ma et al. (2023), according to Eq. (5), the estimated coefficient of $\text{time} \times \text{treat}$ is $\hat{\beta}$:

$$\hat{\beta} = \beta + \gamma \frac{\text{cov}(\text{time} \times \text{treat} | W)}{\text{var}(\text{time} \times \text{treat} | W)} \quad (5)$$

Where W is all other control variables and fixed effects. γ is the influence of non-observed factors on the growth rate of carbon emissions of enterprises in high-polluting industries. If $\gamma = 0$, then the non-observed factors do not affect the estimation results, that is, $\hat{\beta}$ is proved to be unbiased. Since there is no direct verification of whether γ is 0 or not, this study uses an indirect placebo. Its logic is to make the decision on the growth rate of carbon emissions of enterprises in the high-polluting industries randomized, that is, to find a wrong variable 'time' and 'treat' which theoretically will not affect the result variable to replace 'time' and 'treat'. Since 'time' and 'treat' are randomly generated, the effect of the actual LCCP implementation β is equal to 0. Under this premise, if the estimated $\hat{\beta}$ is not 0, it means that γ is not 0, that is, the estimation results in this paper are biased. Specifically, this paper randomly generates a list of enterprises belonging to the pilot area, thus generating an erroneous estimate $\hat{\beta}$, and estimates model (1) again after 500 random samples, as shown in Fig. 2. It is evident that the regression coefficients of the virtual LCCP policy on the carbon emission growth rate of enterprises in high-polluting industries are mainly distributed around 0 in a normal distribution, and most of them are insignificant. At the same time, the estimated coefficients obviously belong to the outliers, demonstrating that the estimation results are robust.

5.2.3. Propensity score matching (PSM)

The PSM-DID approach was used to conduct the robustness test in order to mitigate any potential sample selectivity bias. Nearest neighbor matching, radius matching, and kernel matching methods were respectively used to estimate PSM-DID and the outcomes are displayed in Table 4. It shows that the estimated coefficients of the dummy variables of the LCCP policy are significantly negative, indicating that the LCCP policy can help enterprises in high-polluting industries slow down the growth rate of carbon emissions, which further proves the robustness of the conclusions.

5.2.4. Excluding other policy effects

In the process of estimating the impact of the LCCP policy on the growth rate of carbon emissions of enterprises in high-polluting industries, there may be other policies that will have an impact on the

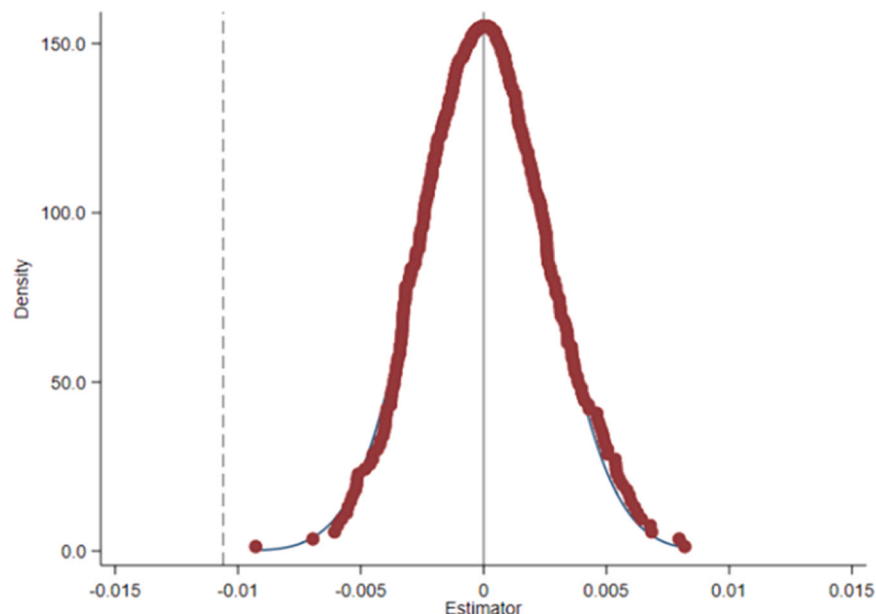


Fig. 2. Placebo test.

Table 4
PSM-DID regression result.

Variable	Nearest neighbor matching	Radius matching	Kernel matching
time×treat	-0.0109** (0.0050)	-0.0104** (0.0050)	-0.0109** (0.0050)
Cons	-0.0111 (0.0684)	0.0322 (0.0672)	-0.0214 (0.0709)
Control	Yes	Yes	Yes
Enterprise FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	2112	2112	2112
R ²	0.0369	0.0369	0.0370

Note: *** denotes $p < 0.01$, ** denotes $p < 0.05$, * denotes $p < 0.1$; enterprise-level clustering robust standard errors in parentheses.

growth rate of carbon emissions of enterprises, resulting in a bias in policy effects. This paper concludes that the policies that may affect carbon emissions from 2010 to 2020 are the special emission limit value policy for air pollutants implemented since 2013, the pilot carbon emissions trading policy in 2011, and the official PM2.5 monitoring policy implemented since 2012. This conclusion is based on a thorough analysis of the factors affecting the growth rate of carbon emissions and other related policies. Therefore, additional policy dummy variables are included in the model (1) for regression in this research. In Table 5, columns (1)–(3) each display the results after excluding the aforementioned policy interference, and column (4) shows the estimation results by simultaneously excluding the aforementioned three policy disturbances. As can be seen, the regression result is still strongly negative and resembles the baseline regression despite the inclusion of the policy dummy factors.

5.2.5. Sample data screening

To prevent extreme values from influencing the baseline regression results, the growth rate of dependent variable carbon emissions was treated by double-tailed treatment on the upper quantile of 1 % and 5 %, respectively, and the regression of Eq. (1) was performed again. The outcomes are displayed in columns (1) and (2) of Table 6. After eliminating extreme values, the estimation results reveal that the finding is comparable to the baseline estimation. Further, the sample is excluded from the four municipalities of Beijing, Tianjin, Shanghai, and Chongqing to control for the bias of special samples on the results. The outcomes are presented in column (3) of Table 6 after regressing Eq. (1) again. The robustness test results are essentially in line with those of the prior benchmark regression.

5.2.6. Independent variable lagged one period

Considering the differences in regional development, the LCCP policy has been specifically adjusted according to the actual situation of

Table 5
Regression results excluding other policy interference.

Variable	(1)	(2)	(3)	(4)
time×treat	-0.0097** (0.0047)	-0.0116** (0.0049)	-0.0106** (0.0047)	-0.0107** (0.0049)
Air Pollutant Policy	Yes	No	No	Yes
Carbon emissions trading policy	No	Yes	No	Yes
PM2.5 monitoring policy	No	No	Yes	Yes
Control	Yes	Yes	Yes	Yes
Enterprise FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Cons	0.0244 (0.0944)	0.0210 (0.0944)	0.0280 (0.0938)	0.0156 (0.0945)
N	2112	2112	2112	2112
R ²	0.1566	0.1566	0.1560	0.1576

Note: *** denotes $p < 0.01$, ** denotes $p < 0.05$, * denotes $p < 0.1$; enterprise-level clustering robust standard errors in parentheses.

Table 6
Other robustness tests.

Variable	(1) 1 % tail reduction	(2) 5 % tail reduction	(3) Delete municipalities	(4) Independent variable lagged one period
time×treat	-0.0075** (0.0034)	-0.0053* (0.0027)	-0.0095* (0.0050)	-0.0087* (0.0051)
Cons	-0.0191 (0.0751)	-0.0208 (0.0557)	0.0477 (0.1037)	0.0808 (0.1070)
Control	Yes	Yes	Yes	Yes
Enterprise FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	2112	2112	1873	1955
R ²	0.1587	0.1553	0.1526	0.1570

Note: *** denotes $p < 0.01$, ** denotes $p < 0.05$, * denotes $p < 0.1$; enterprise-level clustering robust standard errors in parentheses.

each city, and the implementation of the policy may be a delay. As a result, the independent variables in this study are lagged by one period, and the outcomes are displayed in column (4) of Table 6. The fact that the estimated coefficients of the pilot policy's dummy variables remain significantly negative, demonstrates the validity of the paper's regression findings.

5.3. Mechanism analysis

The above research results demonstrate that implementing the LCCP policy has a slowing effect on the growth of carbon emissions. Based on the theoretical analysis above, the mechanism of the LCCP policy to reduce the growth rate of carbon emissions of enterprises lies in the mechanism of technological innovation and environmental protection investment mechanism. This paper examines the mechanism of the LCCP and carbon emission with reference to the two-step method of Jiang (2022). First, based on the theoretical analysis to find mechanism M, there is a relatively direct causal relationship between mechanism M and the explanatory variable Y, which conforms to existing theories and studies. Therefore, it is not necessary to empirically test the causal relationship between M and Y. Secondly, the relationship between independent variable X and dependent variable Y and between independent variable X and mechanism variable M is empirically tested. The specific process is shown in Fig. 3.

5.3.1. Technological innovation in enterprises

The LCCP policy reduces the growth rate of carbon emissions of enterprises in high-polluting industries through technological innovation. This research assesses enterprise technological innovation in terms of two dimensions: R&D capitalization intensity and the number of green patent applications, in order to test the validity of this transmission mechanism. R&D capitalization intensity is a measure of enterprise innovation expenditures, which is an important non-financial performance of enterprises when they innovate with key resources, so capitalization intensity is chosen as the mechanism variable. This paper simultaneously selected green patent data for empirical analysis to more accurately assess the quantity and quality of the output of technological innovation activities of enterprises, according to the "Green List of International Patent Classification". The estimation results of R&D capitalization intensity as a mechanism variable in column (1) of Table 7 display that the LCCP policy significantly improves the technological innovation level of enterprises in high-polluting industries at the 1 % level. This paper uses the number of green patent applications to represent the investment of enterprises in green technological innovation to thoroughly examine the high-quality technological innovation of enterprises. The conclusion is consistent with the previous paper. This may be due to the fact that environmental policy, as regulatory pressure, adds an additional burden on enterprises, thus stimulating them to



Fig. 3. Illustration of the "two-step approach".

Table 7

Regression results of enterprise technological innovation mechanism and green investment mechanism.

Variable	(1) R&D capitalization intensity	(2) The number of green patent applications	(3) Green investment
time×treat	7.2298*** (2.1071)	1.0338* (0.5547)	0.0085** (0.0040)
Cons	5.7891 (78.4541)	8.1442 (6.6457)	0.0752 (0.1246)
Control	Yes	Yes	Yes
Enterprise FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	2112	2112	2112
R ²	0.7702	0.7522	0.4655

Note: *** denotes $p < 0.01$, ** denotes $p < 0.05$, * denotes $p < 0.1$; enterprise-level clustering robust standard errors in parentheses.

innovate and increase their investment in the capitalization ratio of R&D. The regression results in this paper agree with the findings of existing studies (Xu and Cui, 2020; Xiong et al., 2020) that the construction of low-carbon cities induces a rise in the level of green technological innovation of enterprises among high-polluting industries in pilot cities.

Academics have generally confirmed the negative correlation between technological innovation and carbon emissions across the existing studies (Cai et al., 2021; Xia et al., 2024; Li, 2020; Xu et al., 2021). Li (2020) takes listed companies in high-polluting industries as samples and finds that the enhancement of green technology innovation has a positive impact on carbon emission performance and contributes to the advancement of low-carbon development in specific regions. Based on the provincial panel data from 2006 to 2019, Cai et al., (2021) discovered that green technology innovation is negatively correlated with carbon emissions and has a strong spatial spillover effect. Based on the panel data of prefecture-level cities, Xu et al. (2021) found that green innovation has a positive impact on China's carbon emission performance, and the effect of energy consumption structure, industrial structure, urbanization effect and foreign direct investment effect play an intermediary role. Xia et al. (2024) based on 28 provinces and cities as research objects, found that green technology innovation indirectly contributes to the reduction of carbon emissions by promoting the improvement of energy efficiency. And there is significant regional heterogeneity, with the greatest influence on the eastern region. Thus, it has been effectively proved that the LCCP policy influences enterprises' rates of growth in carbon emissions in high-polluting industries through pathways of technological innovation.

5.3.2. Enterprise environmental protection investments

This research draws on the experience of Zhao and Wang (2022) to test the validity of the transmission path that the LCCP policy curbs the growth rate of carbon emissions of enterprises in high-polluting industries by increasing environmental protection investment. According to the definition of green investment, enterprises' green investment expenditure data in the current year is divided by the total assets at the end of the period for standardization, which is used as an indicator of enterprises' environmental investment. According to the findings, which

are displayed in column (3) of Table 7, the estimated coefficient of time×treat is 0.0085, indicating statistical significance. This finding suggests that the LCCP policy greatly enhances the green investment of enterprises. It is clear that enterprises are stepping up their green investment in an effort to lessen pollution. The regression results align with previous studies (Xiao et al., 2023; Zheng and Chang, 2023). The LCCP is a long-term exploratory policy with central-local integration and inter-governmental competition, according to Zheng and Chang (2023), and enterprises tend to make environmental investments to shape a good government-enterprise relationship. According to Xiao et al. (2023), the LCCP policy greatly improved enterprises' environmental investment and strengthened the management of enterprise environmental information disclosure.

Previous research has consistently shown that the increase in environmental investment is favorable to carbon emission reduction (Lee and Min, 2015; Zhang and Li, 2016; Cong et al., 2022; Shen et al., 2021). A study by Lee and Min (2015) on the Japanese manufacturing industry showed that investments in environmental protection can lower carbon emissions and improve enterprise financial performance. The findings of Zhang and Li (2016) align with the conclusion that environmental investment is to improve environmental quality and prevent further ecological degradation and that investment in environmental protection lowers carbon emissions. According to Cong et al. (2022), every 1 % increase in environmental investment results in a 0.558 % decrease in carbon emission intensity. Shen et al. (2021) adopted the autoregressive distributed lag method with cross-sectional augmentation to confirm the negative correlation between green investment and carbon emissions. The research mentioned above demonstrates that the mechanism of enterprise environmental investment is sound, that is, by promoting the level of enterprise environmental investment, the LCCP policy can slow down the growth rate of carbon emissions of enterprises in high-polluting industries.

6. Further analyze

6.1. Heterogeneity tests

6.1.1. Heterogeneity of enterprise ownership

This study examines whether state-owned and non-state-owned enterprises behave differently regarding LCCP policy. Columns (1) and (2)

Table 8

Regression results of heterogeneity analysis.

Variable	(1) State-owned	(2) Non-state-owned	(3) New enterprise	(4) Old enterprise
time×treat	-0.0155** (0.0066)	-0.0038 (0.0065)	-0.0144* (0.0079)	-0.0175 (0.0117)
Cons	-0.0136 (0.0877)	-0.0370 (0.1764)	-0.1190 (0.1648)	-0.4229 (0.4839)
Control	Yes	Yes	Yes	Yes
Enterprise FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	1325	787	813	1299
R ²	0.1726	0.1901	0.2446	0.1584

Note: *** denotes $p < 0.01$, ** denotes $p < 0.05$, * denotes $p < 0.1$; enterprise-level clustering robust standard errors in parentheses.

of Table 8 report the regression results. The estimated coefficients are significant at the 5 % level in the state-owned sample but are smaller and not statistically significant in the non-state-owned sample. It indicates that state-owned enterprises had more prominent carbon emission reduction effects following the LCCP policy when compared to non-state-owned enterprises. On the other hand, state-owned enterprises are subject to stricter environmental protection regulations and are more likely to be constrained by the government. Meanwhile, state-owned enterprises are increasingly focused on the national low-carbon policy, and they will bear more social responsibility in carbon emission reduction.

6.1.2. Heterogeneity of enterprise age

The heterogeneity of enterprise age can lead to different effects in implementing the LCCP policy. For this reason, this research takes the median age of enterprises in the sample as the dividing line and considers enterprises below the median (18 years) as new enterprises. The outcomes, as displayed in columns (3) and (4) of Table 8, reveal that new enterprises reduce the growth rate of carbon emissions, but the effect of old enterprises is not reflected under the LCCP policy. This may be because new enterprises are inclined to place a higher priority on technological innovation and R&D, as well as to incorporate environmental protection measures into all aspects of enterprise operations and create goods in a more ecologically friendly manner. However, older enterprises often have more complicated supply chains, a longer history, and a harder time changing existing procedures or equipment, which makes it difficult for them to implement effective mitigation measures.

6.1.3. Heterogeneity of enterprise scale

To further explore the effects of the LCCP on the growth rate of carbon emissions of enterprises of different scales, this research refers to Ren et al. (2020). Enterprises with total assets larger than the sample mean are classified as large-scale enterprises, and the rest are categorized as small and medium-scale enterprises. The regression results are displayed in columns (1) and (2) of Table 9. The estimated coefficients on the growth rate of carbon emissions in the sample of small and medium-scale enterprises are significantly negative at the 5 % level, indicating that the LCCP effectively reduces the growth rate of carbon emissions of small and medium-scale enterprises. Furthermore, there has been no reflection on the impact of large-scale enterprises on reducing carbon emissions. The estimation outcomes imply that small and medium-sized enterprises are more significantly impacted by the LCCP pilot. This might be because small and medium-sized enterprises are typically more adaptable than large enterprises. They can adapt to market changes and new environmental policies more quickly, thus adopting carbon reduction measures more quickly. Furthermore, small and medium-scale enterprises usually have less capital and resources, so they pay more attention to saving energy and reducing waste.

Table 9
Regression results of heterogeneity analysis.

Variable	(1) Small and medium- scale	(2) Large- scale	(3) High environmental enforcement	(4) Weak environmental enforcement
time×treat	-0.0118** (0.0056)	-0.0022 (0.0072)	-0.0100 (0.0062)	-0.0141* (0.0072)
Cons	-0.0411 (0.0882)	0.0128 (0.1292)	0.0323 (0.1198)	-0.0077 (0.1275)
Control	Yes	Yes	Yes	Yes
Enterprise	Yes	Yes	Yes	Yes
FE				
Year FE	Yes	Yes	Yes	Yes
N	1632	480	1380	732
R ²	0.1632	0.2506	0.1600	0.1661

Note: *** denotes $p < 0.01$, ** denotes $p < 0.05$, * denotes $p < 0.1$; enterprise-level clustering robust standard errors in parentheses.

6.1.4. Heterogeneity of environmental enforcement efforts

In this research, we used the number of cases of environmental administrative penalties in each province to measure the level of environmental law enforcement in each province, and divided them according to the number of cases in each province in the year prior to the implementation of the LCCP (2009). The provinces with low environmental enforcement efforts were those with fewer instances than or equal to the median (1153 cases), and the provinces with high environmental enforcement efforts were those with more cases than that. As displayed in columns (3) and (4) of Table 9, the outcomes indicate that carbon emission reduction in regions with high environmental enforcement is not as fast as those with weak environmental enforcement. And that the LCCP has slowed down the growth of carbon emissions only in areas with weak environmental enforcement. On the one hand, the government may provide more assistance and incentives to enterprises in high-polluting industries in areas with weaker environmental enforcement. Under the impetus of the LCCP, enterprises will actively introduce more advanced and cleaner energy technologies, which may have a more prominent effect on carbon emission reduction. On the other hand, it is possible that enterprises in high-polluting industries in provinces with higher environmental enforcement may use more fossil fuels, and their default costs are higher, making it easier for the enterprises to restrain themselves, thus leading to slower carbon emission reduction.

6.2. Analysis of economic effects

Since the "dual carbon" goal was proposed in 2020, China has taken the initiative to assume responsibility for addressing global climate change and adopted more vigorous measures. As the main body of carbon emissions, enterprises, especially those in high-polluting industries, their response to the LCCP directly affects the effectiveness of the implementation of the policy. The LCCP policy is essential to reducing carbon emissions of enterprises in high-polluting industries. So, can the enterprise's economic efficiency be increased concurrently to create a situation where the economy and the environment both benefit? Based on the analysis above, we continue to explore its impact on the economic effect of enterprises.

Emission reduction initiatives may require enterprises to upgrade technology and modernize equipment, which may need larger investments. This could have an immediate negative impact on the enterprise economy. Some high-carbon industries might face challenges like dwindling market demand, increasing expenses, and heightened competition. But in the long run, reducing carbon emissions can drive enterprises to transition to a greener and more sustainable model. By adopting cleaner energy sources, improving production techniques, and increasing energy efficiency, enterprises can reduce their energy costs and dependence on finite resources. In addition, reducing carbon emissions can help improve enterprise images and reputation, attracting more environmentally conscious consumers and investors. The positive outlook will inevitably drive the enterprise's economic performance and future development. Because of this, this paper adopts ROA and net operating margin to measure the economic effect of the enterprise. The outcomes are displayed in Table 10, where both ROA and net operating margin are significant at the 5 % level, verifying the above analysis. Thus, it validates the previous conjecture that high-polluting industrial enterprises can benefit economically and environmentally from the implementation of the LCCP.

6.3. Analysis of industry spillover effects

The government and all facets of society have been widely concerned about the carbon emission behaviors of those enterprises with high carbon emissions, such as those in the iron and steel, pharmaceutical, textile, mining, metallurgy, coal, chemicals, and other 16 categories of industries defined as high-polluting industries. Under the role of the

Table 10
Analysis of economic effects.

Variable	(1) ROA	(2) net operating margin	(3) R
time×treat	0.0137** (0.0060)	0.0250** (0.0104)	0.0004 (0.0031)
Cons	0.2623*** (0.0740)	0.3510*** (0.1207)	0.0358 (0.0406)
Control	Yes	Yes	Yes
Enterprise FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	2112	2112	4102
R ²	0.5755	0.5218	0.1615

Note: *** denotes $p < 0.01$, ** denotes $p < 0.05$, * denotes $p < 0.1$; enterprise-level clustering robust standard errors in parentheses.

LCCP, the high-polluting industries have put forth a lot of effort to achieve emission reduction targets by adopting technological innovation and increasing investment in environmental protection. By fulfilling its social responsibilities, it has increased public recognition of its environmental performance, which may lead to imitation by enterprises in other industries, thus generating a spillover effect of the LCCP policy. To further examine this issue, this paper refers to [Fang and Tang \(2022\)](#). Every enterprise in the pilot area serves as the treatment group, while every enterprise in the non-pilot area serves as the control group. The outcomes are displayed in column (3) of [Table 10](#). The regression coefficient is not significant, suggesting that the LCCP has no industry spillover effect on the slowdown of carbon emission growth rate while proving the reasonableness of the previous empirical evidence for grouping.

7. Conclusions and policy implications

7.1. Research conclusions

As an environmental regulation to achieve both environmental and economic benefits, the LCCP is in line with the green transformation of enterprises. Based on the sample data of enterprises in China's high-polluting industry from 2010 to 2020, this paper empirically investigates the impact of the LCCP policy on the growth rate of carbon emissions of enterprises in high-polluting industries from the micro-enterprise level by taking a multi-period difference-in-differences model. A series of robustness, heterogeneity and moderating effects tests were conducted. The outcomes display that the LCCP significantly lowers the growth rate of enterprise carbon emissions by an average of 1.06 percentage points. This conclusion is still valid after robustness tests such as parallel trend test, placebo test, PSM-DID, and exclusion of other policy interference. Financing constraints play a negative moderating role between the LCCP and the growth rate of carbon emissions. The analysis of the mechanism shows that the LCCP policy lowers the growth rate of carbon emissions of enterprises in high-polluting industries through technological innovation and increasing environmental protection investment. The heterogeneity study demonstrates that the promotion effect of the LCCP policy on carbon emission reduction of enterprises in high-polluting industries is more obvious among state-owned enterprises, new enterprises, and small and medium-scale enterprises than among non-state-owned enterprises, old enterprises, and large-scale enterprises. And enterprises in areas with weak environmental enforcement are more sensitive to the LCCP policy. Further research finds that there are economic spillovers but not industry spillovers from the LCCP policy.

7.2. Policy implications

By analyzing the impact of the implementation of the LCCP on corporate carbon emissions and exploring the mechanism of its

influence on corporate behavior as well as its path of action, it provides valuable experience and reference for other cities. Continuously summarizing and sharing successful experiences and practices can help more cities make breakthroughs and progress in sustainable development and smart construction inventions, which is conducive to the further promotion and improvement of national policies as well as the implementation of supporting policies. The findings of this paper provide policy insights for the further implementation of the LCCP and the reduction of corporate carbon emissions:

First, formulate appropriate laws and regulations, so that enterprises are required to test carbon emissions on a regular basis and to report and disclose the results. This will guarantee that enterprises are aware of their carbon emissions and formulate corresponding emission reduction measures. For non-state-owned enterprises, large-scale enterprises, and enterprises that have been established for a long time in highly polluting industries, more stringent monitoring requirements can be adopted, such as requiring the implementation of regular third-party audits to improve the accuracy and credibility of the data. Simultaneously, the government may require enterprises to publicly disclose their carbon emissions data to increase transparency and give the public and stakeholders a clearer understanding of their carbon emissions.

Second, the Chinese government should promote manufacturing enterprises, especially those in high-polluting industries, to reduce pollution and carbon through corresponding policy guidance and formulation of procurement strategies. On the one hand, enterprises are urged to adopt the idea of green development and incorporate it into national development strategies and plans. It provides policy guidance in various areas, ranging from industrial restructuring, energy management, and resource utilization to ecological protection, and prompts enterprises to focus on environmentally friendly products and technologies in their research development and innovation by providing guidance and incentives. On the other hand, the government could implement a procurement strategy that favors low-carbon products and services. It acts as a role model in the market, encouraging enterprises to increase their research, development, and production of eco-friendly items by increasing the demand for environmentally friendly products. This tilt not only helps low-carbon enterprises to increase their sales and market share but also promotes the low-carbon transformation of the entire supply chain.

Third, the government encourages R&D and innovation through a series of incentives, exemptions, and subsidies. On the one hand, in order to encourage enterprises to adopt sustainable enterprise practices and motivate them to increase their investment in environmentally friendly goods and services, financial support and tax incentives ought to be given to those engaged in the research and application of low-carbon technologies. Enterprises are assisted in implementing energy-saving and emission reduction technologies, equipment upgrades, and the application of renewable energy. On the other hand, the government can work with financial institutions to create a green financial system and offer specialized financing and credit products to alleviate their financing constraints. Through the establishment of a green loan fund, low-interest loans and risk compensation measures are provided to enterprises that meet environmental protection standards so as to facilitate the financing of sustainable development projects. Furthermore, special funds or subsidy schemes are set up to support enterprises in conducting research, development, and innovation activities related to carbon emission reduction. These funds can be used to research and develop new technologies, products, and processes to help enterprises make strides in reducing carbon emissions.

7.3. Limitations and future exploration

This paper analyzes and verifies the impact of the LCCP on carbon emissions of high-polluting enterprises, but there are still some limitations. First, this paper takes high-polluting enterprises as a sample, and the research can be further expanded to the carbon emissions of

enterprises in other industries in the future. Secondly, this paper only explores the impact of the LCCP, a market-incentivized environmental regulation tool, on the carbon emissions of enterprises. In the future, mandatory environmental regulation tools can be added to comprehensively analyze the impact of carbon emission reduction of enterprises, so as to provide ideas and inspiration for the low-carbon transformation of enterprises. Finally, the current measurement methods of variables and related data collection methods are not comprehensive enough to measure the carbon emission reduction of enterprises. In the follow-up study, it is hoped that carbon emissions can be measured from multiple dimensions, and a more relevant environmental assessment index system can be established to comprehensively reflect the environmental problems of enterprises.

CRediT authorship contribution statement

Wei Sun: Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Jiayu Shen:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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